

Bachelor of Computer Applications (BCS/BCA)

BCS-012: Basic Mathematics

Term-End Examination: June 2025

Full Solved Question Paper in LaTeX

Step-by-step solutions, required diagrams/graphs, and final theory/formula sheet.

Time: 3 Hours **Maximum Marks:** 100

Note: Question No. 1 is compulsory. Attempt any three questions from the remaining questions.

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Question 1

Question. (a) Show that

$$\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$$

Solution. Let

$$D_1 = \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix}, \quad D_2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$$

Both are circulant-type determinants. Expanding the first determinant gives

$$D_1 = -2(a+b+c)(a^2+b^2+c^2-ab-bc-ca).$$

Similarly,

$$D_2 = -(a+b+c)(a^2+b^2+c^2-ab-bc-ca).$$

Therefore,

$$D_1 = 2D_2.$$

Hence proved.

Question. (b) If

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix},$$

find $f(A)$, where $f(x) = x^2 - 5x + 6$.

Solution. Here

$$f(A) = A^2 - 5A + 6I_3.$$

First,

$$A^2 = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}.$$

Also,

$$-5A = \begin{bmatrix} -10 & 0 & -5 \\ -10 & -5 & -15 \\ -5 & 5 & 0 \end{bmatrix}, \quad 6I_3 = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}.$$

Thus

$$f(A) = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} + \begin{bmatrix} -10 & 0 & -5 \\ -10 & -5 & -15 \\ -5 & 5 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}.$$

$$f(A) = \begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$$

Question. (c) Use the principle of mathematical induction to show that

$$1 + 2 + 2^2 + \dots + 2^{n-1} = 2^n - 1, \quad n \in \mathbb{N}.$$

Solution. Let $P(n)$ be the statement

$$1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1.$$

For $n = 1$, LHS = 1 and RHS = $2^1 - 1 = 1$. So $P(1)$ is true.

Assume $P(k)$ is true:

$$1 + 2 + 2^2 + \cdots + 2^{k-1} = 2^k - 1.$$

Then

$$1 + 2 + \cdots + 2^{k-1} + 2^k = (2^k - 1) + 2^k = 2 \cdot 2^k - 1 = 2^{k+1} - 1.$$

Hence $P(k+1)$ is true. Therefore, by mathematical induction,

$$1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1 \quad \forall n \in \mathbb{N}.$$

Question. (d) If $|z - 1| = |z - i|$, show that $\operatorname{Re}(z) = \operatorname{Im}(z)$.

Solution. Let

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

Then

$$|z - 1| = |(x - 1) + iy| = \sqrt{(x - 1)^2 + y^2},$$

and

$$|z - i| = |x + i(y - 1)| = \sqrt{x^2 + (y - 1)^2}.$$

Given $|z - 1| = |z - i|$, so

$$(x - 1)^2 + y^2 = x^2 + (y - 1)^2.$$

Expanding,

$$x^2 - 2x + 1 + y^2 = x^2 + y^2 - 2y + 1.$$

Thus

$$-2x = -2y \Rightarrow x = y.$$

Therefore,

$$\operatorname{Re}(z) = x = y = \operatorname{Im}(z).$$

Question. (e) Find the sum to n terms of the series

$$9 + 99 + 999 + \cdots.$$

Solution. The r -th term is

$$9, 99, 999, \dots = 10^r - 1.$$

Therefore,

$$S_n = \sum_{r=1}^n (10^r - 1) = \sum_{r=1}^n 10^r - n.$$

Now

$$\sum_{r=1}^n 10^r = 10 + 10^2 + \cdots + 10^n = \frac{10(10^n - 1)}{9}.$$

Hence

$$S_n = \frac{10(10^n - 1)}{9} - n.$$

Equivalently,

$$S_n = \frac{10^{n+1} - 10 - 9n}{9}.$$

Question. (f) If α and β are roots of $2x^2 - 8x - 5 = 0$, find a quadratic equation whose roots are α^2 and β^2 .

Solution. For

$$2x^2 - 8x - 5 = 0,$$

we have

$$\alpha + \beta = \frac{8}{2} = 4, \quad \alpha\beta = \frac{-5}{2}.$$

The new roots are α^2 and β^2 . Their sum is

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 4^2 - 2\left(-\frac{5}{2}\right) = 16 + 5 = 21.$$

Their product is

$$\alpha^2\beta^2 = (\alpha\beta)^2 = \left(-\frac{5}{2}\right)^2 = \frac{25}{4}.$$

Thus the required equation is

$$x^2 - 21x + \frac{25}{4} = 0.$$

Multiplying by 4,

$$\boxed{4x^2 - 84x + 25 = 0}$$

Question. (g) Kiaan wants to buy some colour boxes and books to donate to an orphanage. He wishes to buy at least 4 books and 4 colour boxes. A colour box costs Rs. 200 whereas a book costs Rs. 400. How many colour boxes and books should he buy so that expenditure does not exceed Rs. 4,000 and at the same time he can buy maximum number of items?

Solution. Let

$$x = \text{number of colour boxes}, \quad y = \text{number of books}.$$

The conditions are

$$x \geq 4, \quad y \geq 4, \quad 200x + 400y \leq 4000.$$

Dividing the budget inequality by 200,

$$x + 2y \leq 20.$$

The objective is to maximize the total number of items:

$$Z = x + y.$$

Since a colour box costs Rs. 200 and a book costs Rs. 400, colour boxes are cheaper. To maximize the number of items, take the minimum required number of books:

$$y = 4.$$

Then

$$x + 2(4) \leq 20 \Rightarrow x \leq 12.$$

Hence maximum value is obtained at

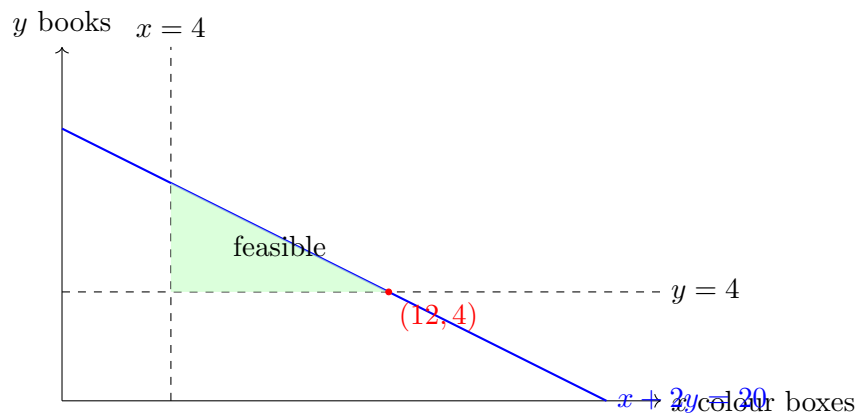
$$x = 12, \quad y = 4.$$

So he should buy 12 colour boxes and 4 books. The expenditure is

$$12(200) + 4(400) = 2400 + 1600 = 4000.$$

The maximum number of items is

$$12 + 4 = 16.$$



12 colour boxes and 4 books; maximum items = 16

Question 2

Question. (a) Using elementary row operations, find inverse of

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Solution. We augment A with I_3 :

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{array} \right].$$

Applying elementary row operations and reducing the left side to identity gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{3}{5} & \frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & \frac{2}{5} & -\frac{3}{5} & \frac{2}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \end{array} \right].$$

Therefore,

$$A^{-1} = \begin{bmatrix} -\frac{3}{5} & \frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & -\frac{3}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \end{bmatrix}.$$

Equivalently,

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Question. (b) If $x = 3 + 2i$, find the value of

$$x^4 - 4x^3 + 4x^2 + 8x + 40.$$

Solution. Substitute $x = 3 + 2i$. First,

$$x^2 = (3 + 2i)^2 = 9 + 12i + 4i^2 = 5 + 12i.$$

Then

$$x^3 = x^2x = (5 + 12i)(3 + 2i) = 15 + 10i + 36i + 24i^2 = -9 + 46i.$$

Also

$$x^4 = x^3x = (-9 + 46i)(3 + 2i) = -27 - 18i + 138i + 92i^2 = -119 + 120i.$$

Now

$$\begin{aligned} x^4 - 4x^3 + 4x^2 + 8x + 40 &= (-119 + 120i) - 4(-9 + 46i) + 4(5 + 12i) + 8(3 + 2i) + 40 \\ &= (-119 + 120i) + (36 - 184i) + (20 + 48i) + (24 + 16i) + 40 \\ &= 1 + 0i. \end{aligned}$$

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Question. (c) Find the sum of an infinite G.P. whose first term is 15 and fourth term is $\frac{3}{25}$.

Solution. Let the first term be $a = 15$ and common ratio be r . The fourth term is

$$ar^3 = \frac{3}{25}.$$

Thus

$$15r^3 = \frac{3}{25} \Rightarrow r^3 = \frac{3}{25 \cdot 15} = \frac{1}{125}.$$

Hence

$$r = \frac{1}{5}.$$

The sum of an infinite G.P. is

$$S_{\infty} = \frac{a}{1-r}, \quad |r| < 1.$$

So

$$S_{\infty} = \frac{15}{1 - \frac{1}{5}} = \frac{15}{\frac{4}{5}} = \frac{75}{4}.$$

$\frac{75}{4}$

Question. (d) If α and β are roots of $x^2 - 4x + 5 = 0$, find the quadratic equation whose roots are $\alpha^2 + 3$ and $\beta^2 + 3$.

Solution. For

$$x^2 - 4x + 5 = 0,$$

we have

$$\alpha + \beta = 4, \quad \alpha\beta = 5.$$

New roots are

$$\alpha^2 + 3, \quad \beta^2 + 3.$$

Their sum is

$$(\alpha^2 + 3) + (\beta^2 + 3) = \alpha^2 + \beta^2 + 6.$$

Now

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 16 - 10 = 6.$$

So the sum of new roots is

$$6 + 6 = 12.$$

Their product is

$$(\alpha^2 + 3)(\beta^2 + 3) = \alpha^2\beta^2 + 3(\alpha^2 + \beta^2) + 9.$$

Therefore,

$$= 5^2 + 3(6) + 9 = 25 + 18 + 9 = 52.$$

Hence required equation is

$$\boxed{x^2 - 12x + 52 = 0}$$

Question 3

Question. (a) Find

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{\sqrt{x}}.$$

Solution. Rationalize the numerator:

$$\frac{\sqrt{x+5} - \sqrt{5}}{\sqrt{x}} \cdot \frac{\sqrt{x+5} + \sqrt{5}}{\sqrt{x+5} + \sqrt{5}} = \frac{x}{\sqrt{x}(\sqrt{x+5} + \sqrt{5})}.$$

Thus

$$\frac{\sqrt{x+5} - \sqrt{5}}{\sqrt{x}} = \frac{\sqrt{x}}{\sqrt{x+5} + \sqrt{5}}.$$

Taking $x \rightarrow 0^+$,

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sqrt{x+5} + \sqrt{5}} = \frac{0}{2\sqrt{5}} = 0.$$

$$\boxed{0}$$

Question. (b) If

$$y = [x(x-1)(x+2)]^{5/7},$$

find $\frac{dy}{dx}$.

Solution. Let

$$u = x(x-1)(x+2).$$

Then

$$y = u^{5/7}.$$

Now

$$u = x(x^2 + x - 2) = x^3 + x^2 - 2x,$$

so

$$\frac{du}{dx} = 3x^2 + 2x - 2.$$

By chain rule,

$$\frac{dy}{dx} = \frac{5}{7} u^{-2/7} \frac{du}{dx}.$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{5(3x^2 + 2x - 2)}{7[x(x-1)(x+2)]^{2/7}}}$$

Question. (c) Find the local extrema of

$$f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 + 105.$$

Solution. Differentiate:

$$f'(x) = 3x^3 - 24x^2 + 45x = 3x(x^2 - 8x + 15).$$

Therefore,

$$f'(x) = 3x(x - 3)(x - 5).$$

Critical points are

$$x = 0, 3, 5.$$

Second derivative is

$$f''(x) = 9x^2 - 48x + 45.$$

Now

$$f''(0) = 45 > 0,$$

so $x = 0$ is a local minimum. Also

$$f''(3) = 81 - 144 + 45 = -18 < 0,$$

so $x = 3$ is a local maximum. Finally,

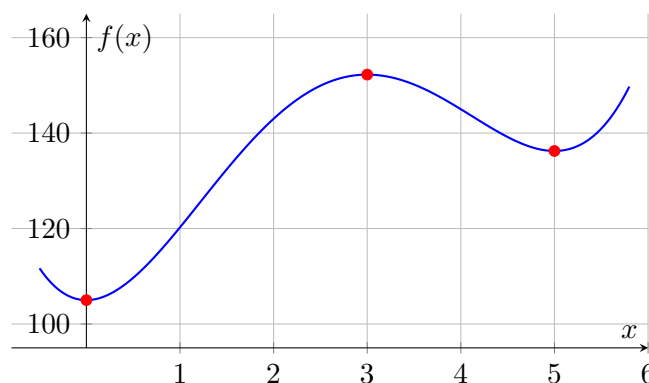
$$f''(5) = 225 - 240 + 45 = 30 > 0,$$

so $x = 5$ is a local minimum.

Function values:

$$f(0) = 105, \quad f(3) = \frac{609}{4}, \quad f(5) = \frac{545}{4}.$$

Local minima: $(0, 105), (5, 545/4)$; local maximum: $(3, 609/4)$



Question. (d) If

$$y = ax + \frac{b}{x},$$

show that

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0.$$

Solution. Given

$$y = ax + bx^{-1}.$$

Differentiate:

$$\frac{dy}{dx} = a - bx^{-2} = a - \frac{b}{x^2}.$$

Again,

$$\frac{d^2y}{dx^2} = 2bx^{-3} = \frac{2b}{x^3}.$$

Now

$$\begin{aligned} x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y &= x^2 \left(\frac{2b}{x^3} \right) + x \left(a - \frac{b}{x^2} \right) - \left(ax + \frac{b}{x} \right) \\ &= \frac{2b}{x} + ax - \frac{b}{x} - ax - \frac{b}{x} \\ &= 0. \end{aligned}$$

Hence proved.

Question 4

Question. (a) Evaluate

$$\int x\sqrt{x+2} \, dx.$$

Solution. Put

$$u = x + 2 \quad \Rightarrow \quad x = u - 2, \quad dx = du.$$

Then

$$\int x\sqrt{x+2} \, dx = \int (u-2)u^{1/2} \, du.$$

So

$$= \int (u^{3/2} - 2u^{1/2}) \, du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C.$$

Substituting $u = x + 2$,

$$\int x\sqrt{x+2} \, dx = \frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$$

Question. (b) Evaluate

$$\int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx.$$

Solution. Let

$$u = 1 - x^2 \quad \Rightarrow \quad du = -2x \, dx.$$

Therefore,

$$x \, dx = -\frac{1}{2} du.$$

When $x = 0$, $u = 1$. When $x = 1$, $u = 0$. Hence

$$\int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx = -\frac{1}{2} \int_1^0 u^{-1/2} \, du = \frac{1}{2} \int_0^1 u^{-1/2} \, du.$$

Thus

$$= \frac{1}{2} \left[2u^{1/2} \right]_0^1 = 1.$$

1

Question. (c) Find the length of the curve $y = 4 + 3x$, from $(0, 4)$ to $(2, 10)$.

Solution. For a curve $y = f(x)$, arc length from $x = a$ to $x = b$ is

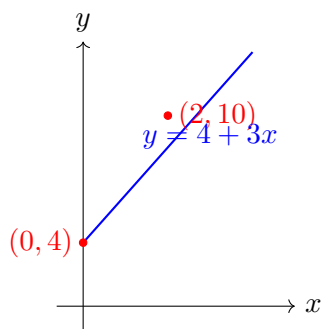
$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Here

$$y = 4 + 3x \quad \Rightarrow \quad \frac{dy}{dx} = 3.$$

From $(0, 4)$ to $(2, 10)$, x runs from 0 to 2. Hence

$$L = \int_0^2 \sqrt{1 + 3^2} dx = \int_0^2 \sqrt{10} dx = 2\sqrt{10}.$$

 $2\sqrt{10}$ 

Question. (d) Show that

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}).$$

Solution. This is the vector product scalar product identity. Using the determinant form of scalar triple products,

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix}.$$

Expanding the determinant,

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}).$$

Hence proved.

Question 5

Question. (a) Find the shortest distance between the lines

$$\frac{x-1}{2} = \frac{y+1}{3} = z$$

and

$$\frac{x+1}{5} = \frac{y-2}{1} = \frac{z-2}{0}.$$

Solution. For the first line,

$$\frac{x-1}{2} = \frac{y+1}{3} = z = t,$$

so a point and direction vector are

$$P_1 = (1, -1, 0), \quad \vec{d}_1 = (2, 3, 1).$$

For the second line,

$$\frac{x+1}{5} = \frac{y-2}{1} = \frac{z-2}{0} = s,$$

so

$$P_2 = (-1, 2, 2), \quad \vec{d}_2 = (5, 1, 0).$$

The shortest distance between two skew lines is

$$D = \frac{|(P_2 - P_1) \cdot (\vec{d}_1 \times \vec{d}_2)|}{|\vec{d}_1 \times \vec{d}_2|}.$$

Now

$$P_2 - P_1 = (-2, 3, 2),$$

and

$$\vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 5 & 1 & 0 \end{vmatrix} = (-1, 5, -13).$$

Thus

$$(P_2 - P_1) \cdot (\vec{d}_1 \times \vec{d}_2) = (-2, 3, 2) \cdot (-1, 5, -13) = 2 + 15 - 26 = -9.$$

So absolute value is 9. Also

$$|\vec{d}_1 \times \vec{d}_2| = \sqrt{(-1)^2 + 5^2 + (-13)^2} = \sqrt{195}.$$

Therefore,

$$D = \frac{9}{\sqrt{195}}$$

Question. (b) Solve the inequality

$$\left| \frac{2x-1}{3} \right| \leq 2.$$

Solution. Using $|A| \leq k \Rightarrow -k \leq A \leq k$, we get

$$-2 \leq \frac{2x-1}{3} \leq 2.$$

Multiplying by 3,

$$-6 \leq 2x-1 \leq 6.$$

Adding 1,

$$-5 \leq 2x \leq 7.$$

Dividing by 2,

$$-\frac{5}{2} \leq x \leq \frac{7}{2}.$$

$$x \in \left[-\frac{5}{2}, \frac{7}{2} \right]$$



Question. (c) Using determinants, show that the points

$$A(a, b + c), \quad B(b, c + a), \quad C(c, a + b)$$

are collinear.

Solution. Three points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are collinear if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Here

$$\Delta = \begin{vmatrix} a & b + c & 1 \\ b & c + a & 1 \\ c & a + b & 1 \end{vmatrix}.$$

Apply the column operation

$$C_2 \rightarrow C_2 + C_1.$$

Then

$$C_2 = \begin{bmatrix} a + b + c \\ a + b + c \\ a + b + c \end{bmatrix}.$$

Thus

$$\Delta = (a + b + c) \begin{vmatrix} a & 1 & 1 \\ b & 1 & 1 \\ c & 1 & 1 \end{vmatrix}.$$

The second and third columns are identical, so the determinant is zero. Therefore,

$$\Delta = 0.$$

Hence the points are collinear.

Question. (d) If

$$y = 2e^x + e^{-x}$$

and

$$\frac{d^2y}{dx^2} = ky,$$

find k .

Solution. Given

$$y = 2e^x + e^{-x}.$$

Differentiate:

$$\frac{dy}{dx} = 2e^x - e^{-x}.$$

Differentiate again:

$$\frac{d^2y}{dx^2} = 2e^x + e^{-x}.$$

But

$$y = 2e^x + e^{-x}.$$

Therefore,

$$\frac{d^2y}{dx^2} = y.$$

Comparing with

$$\frac{d^2y}{dx^2} = ky,$$

we get

$$\boxed{k = 1}$$

Theory and Formula Sheet

Key Theory Required for This Paper

1. Determinants

- If two rows or two columns of a determinant are identical, its value is zero.
- If a row/column operation of the type $R_i \rightarrow R_i + kR_j$ or $C_i \rightarrow C_i + kC_j$ is applied, the determinant does not change.
- Three points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are collinear if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

2. Matrices

- For a polynomial $f(x) = x^2 - 5x + 6$, $f(A) = A^2 - 5A + 6I$.
- To find A^{-1} by elementary row operations, reduce $[A \mid I]$ to $[I \mid A^{-1}]$.

3. Mathematical Induction

- Step 1: Prove the result for $n = 1$.
- Step 2: Assume true for $n = k$.
- Step 3: Prove true for $n = k + 1$.

4. Sequences and Series

- Geometric progression: $a, ar, ar^2, \dots$
- Sum of infinite G.P.: $S_\infty = \frac{a}{1-r}$, where $|r| < 1$.
- $9, 99, 999, \dots$ has r -th term $10^r - 1$.

5. Quadratic Equations

- If roots are α, β , then

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

- Equation with roots p, q is

$$x^2 - (p + q)x + pq = 0.$$

6. Complex Numbers

- If $z = x + iy$, then $\operatorname{Re}(z) = x$ and $\operatorname{Im}(z) = y$.
- $|x + iy| = \sqrt{x^2 + y^2}$.

7. Differentiation

- Chain rule: if $y = [u(x)]^n$, then $\frac{dy}{dx} = n[u(x)]^{n-1}u'(x)$.
- Local extrema occur where $f'(x) = 0$. If $f''(x) > 0$, local minimum; if $f''(x) < 0$, local maximum.

8. Integration and Arc Length

- Substitution method is useful when an expression and its derivative occur together.
- Arc length of $y = f(x)$ from $x = a$ to $x = b$:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

9. Vectors and 3-D Geometry

- Shortest distance between skew lines:

$$D = \frac{|(P_2 - P_1) \cdot (\vec{d}_1 \times \vec{d}_2)|}{|\vec{d}_1 \times \vec{d}_2|}.$$

- Vector identity:

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}).$$

- In a rhombus with adjacent side vectors $\vec{a}, \vec{b}$, diagonals are $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$; their dot product is $|\vec{a}|^2 - |\vec{b}|^2 = 0$.

10. Linear Programming

- Convert word problem into inequalities.
- The maximum or minimum of a linear objective function occurs at a corner point of the feasible region.